Fiscal Year:	FY 2006	**Task Last Updated:**	FY 08/22/2006
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Project Title:	NSCOR: Progressive Alterations of Central Nervous System Structure and Function Are Caused by Charged Particle Radiation		
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Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
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Space Biology Special Category:	None		
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Zip Code:	92350-1700	**Congressional District:**	31
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Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Archambeau, John O. (Loma Linda University) Chang, Polly Y. (SRI International) Fike, John R. (University of California San Francisco) Krucker, Thomas (Novartis Institute for Biomedical Research) Lin, Weili (University of North Carolina at Chapel Hill) Mao, Xao-Wen (Loma Linda University) Obenaus, Andre (Loma Linda University) Pecaut, Michael J (Loma Linda University) Song, Sheng-Kwei (Washington University School of Medicine) Spigelman, Igor (University of California at Los Angeles)		
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Task Description:	The experimental program detailed above will provide a comprehensive, quantitative assessment of the effects of charged particle radiation on the mammalian brain using the mouse hippocampus as a model. The mouse brain will be irradiated with accelerated protons and iron ions and examined over the course of two years for progressive changes in hippocampal structure and function. Cell populations in tissue sections will be quantified by a high precision stereology method to measure rates of loss of neurons, vascular endothelial cells and glia. Brain blood vessel topology will be further characterized by an erosion casting technique that produces a 3-D map of the vessels at 1 micron resolution. Stem cells and ongoing neurogenesis will be quantified using a complementary technique in which dividing cells are directly detected. Activation of cells and residual genetic damage in these cells will be characterized by immunofluorescence and mRNA level measurements of selected cytokine biomarkers and analysis of mutations in the transgene b-galactosidase. Functional impairments of neuronal ensembles will be quantified by electrophysiological measurements of hippocampal tissue slices and patch clamp measurements will detail changes in membrane properties. System level changes will be quantified by electroencephalography and two magnetic resonance imaging modalities that map white matter tracts (diffusion tensor imaging) and local metabolic rate (oxygen extraction). The plasticity of the brain as a system will be tested in its reaction to an immune system challenge mediated by lipopolysacharide. Finally the ability of radiation to modulate the latency of a neurodegeneration process will be tested in a transgenic mouse that overproduced amyloid precursor protein and develops an Alzheimer disease-like pathology. Taken together these results should provide a basis for determining risk to the CNS from charged particle radiation exposure. The main hypothesis for the grant is: Charged particle radiation exposure induces progressive deleterious structural and functional changes in the brain, leading to performance decline. Research Aims: Hypothesis (A) Charged particle irradiation causes a progressive loss of cells and a remodeling of CNS tissue as a function of dose, time and LET. To test hypothesis A we will: Specific Aim A1. Quantify the cellular composition of mouse brain vascular endothelium using stereological analysis of preserved tissue. Specific Aim A2. Quantify the composition of mouse neurons, selected glia and stem cells using stereological analysis of preserved tissue. Specific Aim A3. Quantify the activation of resident and infiltrating mouse immune cells using stereological analysis immune-labeled tissue. Specific Aim A4. Quantify key molecular biomarkers associated with radiation-induced damage to mouse cells and their microenvironment using targeted gene expression profiling and mutation analysis of a structural gene. Hypothesis (B) Charged particle irradiation alters the functional output or performance of the brain as a function of dose, time and LET. To test hypothesis B we will: Specific Aim B1. Quantify tissue-level electrophysiological properties of mouse brain using tissue slices that represent intact regional ensembles of interacting cells. (extracellular, patch clamp) Specific Aim B2. Map the time evolution of macroscopic structural features and metabolic alteration in mouse brains using MRI. Hypothesis (C) Exposure to radiation decreases disease latency and alters response to acute stressors. To test hypothesis C we will: Specific Aim C1. Quantify the magnitude and time course of electrophysiological changes in brain slices of mice expressing a human transgene that produces an Alzheimer disease-like pathology and limits output. Specific Aim C2. Quantify the electrophysiological performance of brain slices in mice reacting to the application of systemic immune stress using lipopolysaccharide as a surrogate infectious agent Foremost in the Critical Path Roadmap system for radiation effects on the central nervous system is Risk #29 – Acute and Late CNS Risks. All of the research and technology questions associated with this risk have priority level one for Mars missions and priority level 3 levels for ISS and 30d lunar missions. The questions addressed by the NSCOR project are 29a (functional changes), 29b (late degeneration), 29c (Alzheimer predisposition), 29e (late degeneration), 29g (cell and tissue mechanisms of damage), 29h (different cell populations including stem cells) and 29i (biomarkers). Also, a NASA workshop on CNS radiobiology identified supporting questions to assist in designing experiments to elucidate CNS effects in response to radiation exposure. These questions include: 1) What is the latency to CNS injury, 2) What cell types are involved in CNS injury (neurons, stem cells, glia)? and 3) What vascular changes occur in response to exposure. We have used this framework to focus our investigation on the effects of charged particles on the CNS. Specifically we will quantify cellular and functional changes in the hippocampus following high energy protons (main SPE constituent) and Fe-56 particles . We will use male rodents with sufficient statistical samples to address inter-individual variation and will address workshop questions 1-3 by examining the responses of different constituent cell types and the latency of damage. NSCOR subproject A quantifies different cell populations (29b, 29e, 29g, 29h); subproject B addresses functional changes and non-invasive techniques for identifying biomarkers (29a, 29i); and subproject C targets late degenerative changes with an Alzheimer model and the role of inflammation (29b, 29c, 29e, 29g and 29i). Our strategy is therefore aligned with the programmatic goals set forth by NASA and its advisory groups.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The response of the central nervous system to ionizing radiation will have direct impact on the development of radiotherapy using charged particles. The ability of protons, helium ions and carbon ions to be localized precisely in treatment fields close to critical structures (eg. optic nerve) make charged particle radiotherapy highly desireable. The limiting biological constraint on such treatments is the reaction of normal tissue surrounding a tumor site. The results of the NSCOR task will quantify cell loss and tissue functional changes form proton and iron irradiation that can be used directly to estimate normal tissue complication probabilities in the context of radiotherapy.
	Administrative Changes. Associate Director, Thomas Krucker, of The Scripps Research Institute (TSRI) took a new position at the Novartis Institute for Biomedical Research in Cambridge, Massachusetts. He will continue as an Associate Director without compensation. This change required a reorganization of numerous activities associated with TSRI and NSCOR hypothesis C. Transgenic APP23 mice have now been transferred to Loma Linda University where they are being bred and genotyped for experimental requirements. Electrophysiological experiments on lipopolysaccharide-challenged animals and APP23 mice have been transferred to Loma Linda with the addition of Dr. Roman Vlkolinsky to the LLU staff from his previous position at TSRI. Finally, arrangements for synchrotron x-ray tomography of vascular corrosion casts have been made directly with Dr. Marco Stampanoni of the Paul Scherrer Institute in Villigen, Switzerland. Technical Progress Irradiations. Most of the animals scheduled for iron irradiation at the Brookhaven National Laboratory have now been treated and are aging to their appropriate harvest dates. Additional groups of transgenic animals are scheduled for irradiation in the October 2006 run. Essentially all hypothesis B animals have been irradiated with both iron ions and protons. Proton irradiations of animals for stereological analysis under hypothesis A will now be started so that subjects can be harvested on a continuous schedule during project years 3 and 4.

Task Progress:

Core Activities. All tissue processing, stereological and immunohistochemical methods have been continuously improved and standard protocols have been established. Adjustments to several immunofluorescence protocols were made to compensate for the limited stability of certain antibodies and to correct for image quality when tomato lectin is used to label the lumens of microvessels. Vascular corrosion casting training by Thomas Krucker is scheduled for early September after which time processing at LLU and shipment to Paul Scherrer Institute will commence. The central database is operating stably using the Linux operating system and the set of servers has been moved to a more permanent location. Mr. Brad Karain, who is the administrator, will be leaving for graduate school in the fall and his replacement is being sought. Electrophysiology technician Tyra Lamp has also left for a new position and her replacement is being sought. It is anticipated that over the next 12 months the MRI facility currently operated by the Department of Radiation Medicine will become jointly operated with the Department of Radiology. Associate Director Andre Obenaus will continue to oversee operations. Plans are in the works for a project review to be held in northern California this fall.

Hypothesis A Activities. Stereological analysis of iron irradiated mice has been completed for animals up to 12 months of age. The systematic sections cut from each brain slice containing compartments of interest were stained to identify the capillaries, nuclei of endothelial cells as well as neurons and glia. Microvessel endothelial cell population numbers as well as vessel lengths in CA1 and dentate gyrus (DG) regions of the hippocampus 6 and 12 months following 56Fe were measured but no significant reduction of endothelial cell density and capillary density was noted at the 6 month time point. However, the stereological methods were shown to be sensitive and demonstrated that there were highly significant differences in microvessel length density as well as endothelial number between dentate gyrus and hilus or CA1 region of the hippocampus. Previous work with rat retinal vasculature suggests that late degenerative changes in microvasculature will first become manifest at about 15 months. With respect to microglia there appears to be an effect of 4 Gy iron irradiation but the results are not yet statistically significant. Certain methodological setbacks required re-imaging and re-counting of a number of sections because the IBA1 antibody appears to be unstable and the tissues must be sectioned, labeled, and imaged with very little margin for delay. We have since modified the protocols appropriately. Most of the sections have already been re-imaged and we are in the process of counting.

For analysis of neurogenesis, tissues have been processed from 57 mice representing doses of 0 to 4 Gy of iron and follow-up times of 1 and 6 months. Many of the sections have been stained and analyzed. Additional tissues from 28 mice that had received varying doses 2 months previously have also been sectioned and stained. Analyses of the latter animals is underway. Based on the analyses carried out to date, it is clear that 56Fe irradiation induces a significant and persistent decrease in the production of new neurons within the dentate gyrus. Furthermore, the reductions appear to be associated with alterations in the microenvironment, including increased indications of inflammation and oxidative stress.

For assessment of DNA damage and repair, hippocampal tissues were isolated from animals exposed to a range of iron doses at 8 weeks after irradiation and processed for transgene mutant frequencies (MF). The spontaneous MF in the hippocampus is consistent with our historical whole brain values. We observed a trend of increasing MF with increasing radiation dose up to 2 Gy. Analyses of characteristics of mutants obtained from the 8 week samples using restriction fragment length polymorphism (RFLP) are in progress. Pooled results from >50 mutants from the 2 Gy treatment group suggest that > 70% of the mutants contain either point mutations or <1 kb deletions. These results suggest that only cells containing small changes in the genome continue to survive in tissues at 8 weeks post irradiation. Heavily damaged cells containing large genomic damage are most probably eliminated from the surviving cell population at this late time.

Hypothesis B activities. Electrophysiology: Recordings were obtained at 1-3 mo after irradiation. Animals underwent standard in vitro electrophysiological preparation, yielding 400 um thick hippocampal sections. Extracellular recordings were performed in the CA1 dendritic region. After electrophysiological characterization, long term potentiation (LTP) was induced by tetanic stimulation with recordings for 90 mins post LTP induction. Hippocampal LTP was readily induced in 0 Gy animals but this contrasts with head-only irradiated animals at 1 mo after irradiation which had dose-dependent decrements in PTP. Similarly, LTP in 1 mo post-irradiation was decreased at 4 Gy. However, in 3 mo post irradiation animals, there was no significant change between 0 Gy and treated animals. These results demonstrate altered hippocampal plasticity at 1 mo after 56Fe head-only irradiation. At 3 mo these changes appeared to normalize but underlying pathology may lead to other altered hippocampal electrophysiological characteristics.

Magnetic Resonance Imaging : After the transfer of technology involved in small animal diffusion tensor imaging (DTI), the challenge of incorporating non-compatible data format of the MR data in between Washington University and LLU has been addressed. Several custom made computer programs have been created to accept data generated from LLU's system to construct the quantitative DTI maps. Briefly, the required information embedded in the parameter set of Bruker MRI (the MR system used in LLU) data set was extracted and fed to programs written in Matlab for DTI map calculation. A water phantom was tested by acquiring a DTI data set according to our specifications. The homogeneously distributed image intensity in the raw data suggests a negligible level of artifact in the data set. So, we have established the necessary software to handle the DTI data set for analysis. To ensure the quality of DTI on mouse brain, a Washington University animal holder design has been fabricated and tested at LLU and a custom designed surface coil has been adopted for improved image quality.

Cerebral metabolic rate of oxygen utilization (CMRO2) is a critical physiological parameter for insights into tissue viability during acute cerebral ischemia and we hypothesize that it will reveal vasculature changes following irradiation. This parameter has been studied using modified T2 images acquired after middle cerebral artery occlusion (MCAO) in a rat model. The methods are being adapted at University of North Carolina first to mice then to the Brucker MRI system in use at LLU. A number of software components have now been developed for this purpose and implementation on the LLU Brucker is anticipated in the near future.

Hypothesis C Activities. Electrophysiology: In immunologically challenged mice (lipopolysaccharide injection) the most significant finding was the inhibitory effect of radiation on excitability of CA1 neurons. This was detected as a decrease of the amplitude of population spike evoked at maximal stimulation intensity. The effect was apparent 3 month after radiation, became very significant at the dose of 4 Gy six months after radiation and at 12 months after radiation the effect was clearly dose-dependent. This indicates detrimental effect of radiation predominantly on dendro-somatic coupling function that is responsible for generation of the action potential (functional output of CA1 neurons). In APP23 transgenic mice we measured changes of selected electrophysiological parameters (excitability, synaptic efficacy, short-term and long-term synaptic plasticity) at iron radiation doses of 0, 1, 2 and 4 Gy. In 9 month old mice we were able to detect changes in synaptic efficacy following 1Gy exposures. The decrease was even more prominent at 4 Gy.

	Such changes were not observed at the age of 2 and 6 months and at the age of 14 months the decrease was less significant due to the reduction of synaptic efficacy in control, non-irradiated mice, presumably caused by aging. The change in synaptic efficacy may reflect changes in synaptic density and dendritic arborization of CA1 neurons due to radiation exposure. Vascular corrosion casting of transgenic mice will begin in September 2006. The delays were due to switch in location of the APP23 animal colony and the revision of contractual arrangements for synchrotron x-ray tomography.
Bibliography Type:	Description: (Last Updated: 03/26/2026)
Abstracts for Journals and Proceedings	Fike JR, Rola R, Limoli CL, Potts M, Fishman KM, Baure, J. "High LET radiation effects on hippocampal neurogenesis." 4th International Workshop on Space Radiation Research and 17th Annual NASA Space Radiation Health Investigators' Workshop. Moscow – St. Petersburg, June 5 - 9, 2006. 17th Annual NASA Space Radiation Health Investigators' Workshop proceedings, June 2006. , Jun-2006
Abstracts for Journals and Proceedings	Kao J, Mao XW, Obenaus A, Song SK. "Radiation induced injury of the lens and retina in rat eyes: Assessment with diffusion tensor imaging." ISMRM 14th Scientific Meeting. Seattle, Washington, January 2006. ISMRM 14th Scientific Meeting, Jan 2006. , Jan-2006
Abstracts for Journals and Proceedings	Mao X, Archambeau J, Archambeau P, Mekonnen T, Larsen S. "Stereological analysis of dose response of microvessel parameter in brain to iron and proton irradiation: A progress report." 4th International Workshop on Space Radiation Research and 17th Annual NASA Space Radiation Health Investigators' Workshop. Moscow – St. Petersburg, June 5 - 9, 2006. 17th Annual NASA Space Radiation Health Investigators' Workshop proceedings, June 2006. , Jun-2006
Abstracts for Journals and Proceedings	Mao XW, Archambeau P, Baqai W, Archambeau J. "A quantification study of brain vasculature parameter to proton and iron irradiation using stereology methods." 52nd Annual Meeting of Radiation Research Society, Denver, Colorado. October 2005. 52nd Annual Meeting of Radiation Research Society proceedings, Oct 2005. , Oct-2005
Abstracts for Journals and Proceedings	Mao XW, Pecaut MJ, Favre C, Nelson GA. "Quantitation of radiation-induced cell loss and tissue remodeling in mouse hippocampus: a progress report for aim a of the NASA CNS NSCOR project." 4th International Workshop on Space Radiation Research and 17th Annual NASA Space Radiation Health Investigators' Workshop. Moscow – St. Petersburg, June 5 - 9, 2006. 17th Annual NASA Space Radiation Health Investigators' Workshop proceedings, June 2006. , Jun-2006
Abstracts for Journals and Proceedings	Obenaus A, Lamp T, Vlkolinský R, Nelson GA, Favre C, Krucker T. "Alterations in hippocampal electrophysiological characteristics and development of hyper-excitability after 56Fe radiation exposure." 4th International Workshop on Space Radiation Research and 17th Annual NASA Space Radiation Health Investigators' Workshop. Moscow – St. Petersburg, June 5 - 9, 2006. 17th Annual NASA Space Radiation Health Investigators' Workshop proceedings, June 2006. , Jun-2006
Abstracts for Journals and Proceedings	Vlkolinsky R, Lamp T, Nelson GA, Krucker T, Obenaus, A. "Lipopolysaccharide-induced impairment of synaptic plasticity in the mouse hippocampus: Effect of 56Fe radiation." 4th International Workshop on Space Radiation Research and 17th Annual NASA Space Radiation Health Investigators' Workshop. Moscow – St. Petersburg, June 5 - 9, 2006. 17th Annual NASA Space Radiation Health Investigators' Workshop, June 2006. , Jun-2006
Abstracts for Journals and Proceedings	Vlkolinsky R, Lamp T, Obenaus A, Nelson G., Krucker T. "Effect of 56Fe Radiation on Lipopolysaccharide-Induced Impairment of Synaptic Plasticity in Mouse Hippocampus." Society for Neuroscience, Washington, DC, November 2005. Society for Neuroscience, Washington 2005, poster 503.9. , Nov-2005
Articles in Peer-reviewed Journals	Mao X-W. "A quantitative study of the effects of ionizing radiation on endothelial cells and capillary-like network formation." Technol Cancer Res Treat. 2006 Apr;5(2):127-34. PMID: 16551132 , Apr-2006
Articles in Peer-reviewed Journals	Rola R, Sarkissian V, Obenaus A, Nelson GA, Otsuka S, Limoli CL, Fike JR. "High-LET radiation induces inflammation and persistent changes in markers of hippocampal neurogenesis." Radiat Res. 2005 Oct;164(4 Pt 2):556-60. PMID: 16187787 , Oct-2005
Papers from Meeting Proceedings	Fike JR, Rola R, Limoli CL. "High LET Irradiation Induces Long-Term Reductions of Neural Precursor Cells and Their Progeny." 52nd Annual Meeting of the Radiation Research Society, Denver, CO, October 2005. 52nd Annual Meeting of the Radiation Research Society proceedings, Oct 2005. , Oct-2005